

opg 1

skærer y-aksen i : 3

hældning : 1

$$y = 1 \cdot x + 3 \quad \text{ellv} \quad \underline{\underline{y = x + 3}}$$

opg 2

Bestem forskrifter:

$$\text{Linje 1: } A = (2, 5) \quad B = (0, 3)$$

$$\text{Linje 2: } A = (2, 5) \quad C = (7, 0)$$

$$\text{Linje 1: } a = \frac{5-3}{2-0} = \frac{2}{2} = 1, \quad b = 5 - 1 \cdot 2 = 3$$

$$\underline{\underline{y = x + 3}}$$

$$\text{Linje 2: } a = \frac{5-0}{2-7} = \frac{5}{-5} = -1, \quad b = 5 - (-1) \cdot 2 = 7$$

$$\underline{\underline{y = -x + 7}}$$

opg 3

Løsninger fundet vha.

"Lige store koefficienters metode"

$$y = 2 \cdot x + 1 \quad (L1)$$

$$3 \cdot y = -3x + 12 \quad (L2)$$

Jeg ganger $(L1)$ med -3 , og regner $-3 \cdot L1 + L2$:

$$-3 \cdot y = -6 \cdot x - 3 \quad (-3 \cdot L1)$$

$$3 \cdot y = -3 \cdot x + 12 \quad (L2)$$

$$\underline{\underline{0 = -9x + 9}} \quad (-3 \cdot L1 + L2)$$

$$\underline{\underline{x = 1}}$$

Ved indsættelse i $L1$ fåes:

$$y = 2 \cdot 1 + 1 \Leftrightarrow \underline{\underline{y = 3}}$$

OPG 4 Lösminger fundet hva.
Substitution:

$$y = 3x + 1 \quad \wedge \quad 2y = 2x + 8 \quad \Downarrow$$

$$\wedge \quad 2 \cdot (3x + 1) = 2x + 8 \quad \Downarrow$$

$$6x + 2 = 2x + 8 \quad \Downarrow$$

$$4x = 6 \quad \Downarrow$$

$$x = \frac{3}{2}$$

$$\Downarrow \quad y = 3 \cdot \left(\frac{3}{2}\right) + 1 \quad \wedge$$

$$\Downarrow \quad y = \frac{9}{2} + \frac{2}{2}$$

$$\Downarrow \quad \underline{\underline{y = \frac{11}{2}}}$$

OPG 5 FIND NULPUNKTER $f(x) = -4x + 4$

$$\Downarrow \quad 0 = -4x + 4$$

$$\underline{\underline{x = 1}}$$

Nulpunkter (1, 0)

OPG 6 LOS LIGNING

$$\frac{x+2}{x+1} = \frac{2x}{3x}$$

$$\Downarrow \quad \frac{x+2}{x+1} = \frac{2}{3}$$

$$\Downarrow \quad (x+2)3 = 2 \cdot (x+1)$$

$$\Downarrow \quad 3x + 6 = 2x + 2$$

$$\Downarrow \quad \underline{\underline{x = -4}}$$